For your eyes only? An eye-tracking experiment investigating microtargeting transparency, visual attention, and critical processing

Martin-Pieter Jansen[1], and Iris van Ooijen[2]

[1]Social Psychology: Media and Communication, University of Duisburg-Essen, 47057, Duisburg, Germany

[2]Behavioural Science Institute, Radboud University, 6525 GD, Nijmegen, The Netherlands

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Author Note

Martin-Pieter Jansen [https://orcid.org/0000-0003-2876-8073]

Iris van Ooijen [https://orcid.org/0000-0002-1907-6158]

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Correspondence concerning this article should be addressed to Martin-Pieter Jansen, Social Psychology: Media and Communication, University of Duisburg-Essen, Forsthausweg 2, 47057 Duisburg, Germany. E-Mail: martin-pieter.jansen@uni-due.de

Abstract

By means of their personal data, users on social media platforms get targeted to be persuaded by political advertisers with messages that are specifically tailored to them: political microtargeting. A substantial problem with political microtargeting is that receivers of targeted advertisements are often unaware that they are being targeted. Increasing transparency to improve awareness potentially leads to a more critical way to process information in an advertisement. Moreover, the use of these measures could lead to differences in users' attention to the advertisement. Therefore, this study investigates whether a disclosure that informs users about targeting practices occurring on Instagram leads to more visual attention to the disclosure, whether this leads to more critical processing of the advertisement, and whether this relationship is mediated by users' visual attention to the advertisement. In a preregistered one-factorial between-subjects laboratory eye-tracking experiment (N = 134) we exposed participants to an Instagram timeline containing either a sponsorship disclosure (control condition) or a more salient and informative targeting disclosure (experimental condition). Results show that the targeting disclosure did not lead to higher levels of visual attention to the disclosure, but that attention to the disclosure was positively related to more critical processing as well as more visual attention to the ad. Nevertheless, we did not find a mediating effect of attention to the advertisement on the relationship between attention to the disclosure and critical processing. Finally, we believe that these disclosures represent a step forward in improving transparency and shielding users.

Keywords: microtargeting, transparency, eye-tracking, disclosures, political

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Recently, strategies used in political advertising have shifted from canvassing–like door-to-door interactions with potential voters–to online interactions. The majority of communication between voters and politicians occurs online, more specifically on social networking sites (SNS) (Giasson et al., 2019). This shift to an online environment creates opportunities for political advertisers. Online platforms collect the interactions and behavior of their users (e.g., liking posts, commenting on videos, and exposure time to content). This information can be used to build profiles of certain user groups and allocate them into small groups to target them with specific messages (Murray & Scime, 2010). These messages are developed to resonate more effectively within these groups, a practice called *microtargeting*, or in a political context: *political microtargeting* (PMT) (Binder et al., 2022; Kruikemeier et al., 2022; Zuiderveen Borgesius et al., 2018).

The most commonly known example of PMT is when *Cambridge Analytica*, a British consultancy firm, gathered and used data from a large number of Facebook users to form psychological profiles and, based on these profiles, sent target messages to users that would persuade them as much as possible (Cadwalladr, 2018). Hereby, the firm allegedly contributed to Trump's presidential race and the Leave campaign's success in the Brexit referendum. However, PMT also has the potential to benefit political systems within societies. The technique has the potential to, for instance, activate voters who are usually deemed to have a lower propensity to vote by reaching out to them in personally relevant ways with messages that potential voters really care about (Zarouali et al., 2020), and also strengthen general political interest (Matthes et al., 2022) and general political knowledge (Holt et al., 2013).

While at first glance PMT might seem to be an effective strategy in political campaigning, the phenomenon also introduces risks that might not outweigh the benefits. For instance, PMT discourages political participation by showing people more issues that they do not find interesting (Bodó et al., 2017) and decreases scrutiny through the exclusion of certain target groups if they are not interesting for advertisers (Jamieson, 2013). Critics claim that users' privacy is at risk when profiles are developed based on personal information and (online) behavioral data, which occurs mostly without their informed consent (Bennett, 2015; Dobber et al., 2019; Tene, 2011). A major problem with PMT is that it may manipulate voters by targeting them while they are unaware of it (Zuiderveen Borgesius et al., 2018). Users generally do not recognize that they have received persuasive messages that are specifically targeted at them (Binder et al., 2022; Kruikemeier et al., 2016). A potential solution is the use of disclosures.

Disclosures

Disclosures are labels that are usually implemented to inform users about the sponsored or commercial nature of a message (e.g., advertorials and YouTube reviews). Within research on sponsored content, disclosures have been shown to facilitate the recognition of advertisements (Amazeen & Wojdynski, 2020; Boerman et al., 2015). On platforms such as Facebook, Instagram, and Twitter, disclosures are a measure to improve transparency, while simultaneously providing users with tools to distinguish sponsored content from regular content (Boerman & van Reijmersdal, 2016; Janssen et al., 2021).

Regulatory bodies are starting to implement regulations to provide users with disclosures as a transparency measure. A prime example is the European Commission's Digital Service Act (DSA) (European Commission, 2022). The DSA states that targeted messages and ads should include information on when and on whose behalf, content is displayed. Simultaneously, these measures aim to go beyond that and include background

information about the individual data used to target users and parameters for those data points (European Commission, 2022; King, 2022). In the Dutch context, the DSA would be the first time that targeting disclosures have been initiated. However, other regulators have attempted to implement targeting disclosures in their markets. In Brazil, the 'fake news bill', which also includes targeting transparency on platforms, is being reviewed in Congress (Lubianco, 2022). Two examples within the EU are a proposal in Ireland (Online Advertising and Social Media (Transparency) Bill, 2021) and an adapted version in France (Dobber, 2021; Lutte contre la manipulation de l'information, 2018).

One of the main implications of current disclosure research is that the prominence, positioning, and degree to which disclosures stand out, in contrast to regular content on a timeline, are important (Amazeen & Wojdynski, 2020; Wojdynski & Evans, 2016). Moreover, a combination of text and a logo or symbol leads to the highest visual fixation (Boerman et al., 2015). If disclosures are more prominent (i.e., bigger, containing more text, or more colorful), they stand out compared to regular content on a timeline. In turn, they lead to higher levels of attention and clearer perceptions of disclosed information, thus contributing to greater transparency (Binder et al., 2022; Kruikemeier et al., 2016). The currently used disclosures are mostly designed with a grey and thin font (i.e., lacking prominence) that state content is sponsored and by whom it is paid for regarding political ads. However, research shows that these disclosures seem insufficient in informing users and increasing transparency (Binder et al., 2022; Binford et al., 2021; Jansen & Krämer, 2023; Jost et al., 2022).

Nonetheless, since 2020, differently designed disclosures have been used by platforms. During the COVID-19 pandemic, Instagram has implemented measures to counter the spread of misinformation. One of these measures was to place disclosures on posts and stories (i.e., disappearing posts) that informed users that the information could be false or not

yet proven. The disclosure was located below the content and was designed as a bar in the system font and color. Moreover, the disclosure contained a large "I" symbol to show that it contained additional information and a link to the 'COVID-19 Information Center' (Clark, 2020; Instagram, 2020). The design choices made regarding these disclosures might be a solution to the regulations in the DSA. While these disclosures provide platforms with more space to disclose information, they could also attract more attention as they are more prominent. To the best of our knowledge, these specific disclosures have not yet been investigated by other researchers as transparency measures for PMT. Since more legislation and regulations seem to appear concerning ads on SNSs (e.g., the DSA), the aforementioned COVID-19 disclosures have the potential to be a solution for platforms to comply with these regulations.

Visual attention to disclosures

For disclosures to work and contribute to transparency users need to see them first, which is why eye-tracking is deemed a fitting method to investigate the effectiveness of disclosures. Recent work combining an experiment and eye-tracking in the context of disclosures for influencer marketing on Instagram shows that while users seem to be aware of influencer marketing they also make mistakes in identifying it as such within a feed (Boerman & Müller, 2021). Nonetheless, users pay the most attention to paid partnership' labels (compared to no sponsor cue, #paidad or a brand tag). In turn, this influences the recognition of ads through conceptual persuasion knowledge (i.e., ad recognition, understanding of persuasive intent, and understanding of the commercial source) (Boerman & Müller, 2021). Moreover, research on product placement disclosures in television programs showed that a combination of a logo and text was more effective concerning users' recognition of ads than either of those aspects separately (Boerman et al., 2015).

In the field of PMT, research shows that even though they are promising, targeting disclosures do not always get recognized or recalled by users (Binder et al., 2022; Dobber et al., 2023; Jost et al., 2022; Kruikemeier et al., 2016). However, in one of the first eye-tracking studies regarding disclosures on microtargeted ads, Jost and colleagues (2022) found that while a prominent disclosure did not lead to more visual attention to the disclosure, participants were still better able to recall sponsored ads in a timeline. Oppositely, in another eye-tracking experiment, Binford and colleagues (2021) found that although a political ad disclosure on Facebook did not effectively enhance users' comprehension of who paid for the political ad, users did pay visual attention to the disclosure. Therefore, we expect that a targeting disclosure (i.e., a disclosure consisting of a sponsorship element and a targeting element) that provides more information and is more prominent because it consists of more text, leads to more visual attention to the disclosure compared to a sponsorship disclosure (i.e., a disclosure consisting of merely sponsorship elements). This leads to the following hypothesis:

> H1: A more prominent and informative targeting disclosure (compared to a sponsored disclosure) leads to more visual attention towards this disclosure.

Critical processing

One of the constructs that have been identified as important with regard to disclosures and their reception, but is not yet studied in the domain of disclosures and PMT, is critical processing (Boerman & van Reijmersdal, 2016). When users are made aware of a persuasive attempt, they have to figure out how they manage their response to this attempt effectively (Friestad & Wright, 1994). As a result of this, the awareness of that persuasive attempt encourages more systematic and biased processing of the advertisement (Janssen et al., 2010). Critical processing is defined as the adoption of an evaluative style of processing in

which the content is criticized (Boerman et al., 2014, p. 217). This style of processing has been found to get elicited through the use of sponsoring disclosures (compared to no disclosure) in the case of sponsored content on television (Boerman et al., 2014). Moreover, Zarouali and colleagues (2017) showed that adolescents engage in critical reflection of targeted advertising if they are exposed to a cue that triggers their critical elaboration. In other work, Zarouali and colleagues (2018) found that if adolescents get exposed to a disclosure in a timeline regarding the targeted nature of a Facebook ad, they engage in more critical processing compared to a dropdown menu containing that information, or an ad without any transparency measure. When a disclosure enhances the recognition of a post being an ad instead of organic content, users could realize that the post is indeed an ad, not neutral, and has a persuasive purpose. Given the embedded design of the ad, users might even feel deceived. Because of this awareness and feeling of deception, users might cope with the persuasive attempt by criticizing it and being more skeptical while interpreting the message (Boerman et al., 2014). Moreover, awareness of the persuasive intent has been shown to lead to less favorable perceptions of the communicator (Allyn & Festinger, 1961; Boerman et al., 2014; Campbell & Kirmani, 2000). Thus, we expect that visual attention to the disclosure leads to more critical processing, which leads to the following hypothesis:

> H2: Visual attention towards the disclosure is positively related to critical processing of the advertisement.

Visual attention to the advertisement

However, we propose that another mechanism might get activated through visual attention to the disclosure. Disclosures can not only protect users from the persuasive effects of the ad but also have the potential of unintentionally increasing, rather than decreasing the persuasive effects (Janssen et al., 2016). The main reasoning is that when users allocate more

visual attention to a disclosure and read its message, they might become curious about a political party's perception of them. This curiosity, in turn, might lead to an interest in the party's persuasive message. This could mean that attention to the disclosure would lead to more attention to the advertisement which in turn, assuming that the advertisement stays as persuasive as it is intended to be, might be a negative unintended effect.

Prior research shows that, indeed, viewers' recognition of advertising increases their brand memory (Boerman et al., 2015), which could be seen as an outcome of a persuasive message. Moreover, earlier work by Evans and colleagues (2019) shows that sponsorship transparency can have a positive effect on receivers' attitude towards the ad and on receivers' purchase intention. Furthermore, Pan and Zinkhan (2006) found that a privacy disclosure message can lead to more trust in an online retailer. Besides, Zarouali and colleagues (2018) found that when adolescents get exposed to a disclosure in a timeline explaining the targeted nature of a Facebook ad, they perceive the ad as more persuasive. Other work finds that if platforms and/or advertisers are transparent about the personalization practices taking, this decreases perceptions of vulnerability and as a result, consumers are more inclined to engage with advertisements (Aguirre et al., 2015). In this study, we view visual attention to the advertisement as a proxy of users' interest in the advertisement itself, which could eventually lead to the brand –or party– related effects mentioned above. Thus, we expect that users' visual attention to the disclosure relates to more visual attention towards the advertisement itself, leading to the following hypothesis:

> H3: Visual attention towards the disclosure is positively related to visual attention towards the advertisement.

While most research that is mentioned above uses the effectiveness of an advertisement as an outcome measure, we aim to investigate our initial dependent variable,

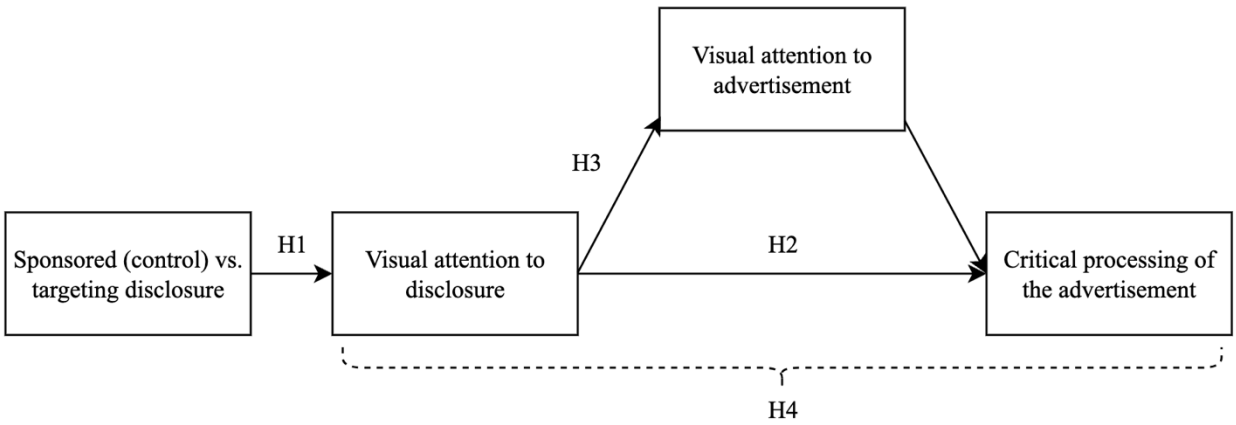
critical processing, as such. Therefore, in this work, we also investigate the potential mediating effect of users' visual attention to the advertisement on the effect of users' visual attention to the disclosure and their critical processing of the ad. This leads to the following hypothesis:

> H4: Visual attention towards the disclosure is positively related to visual attention towards the advertisement which in turn positively relates to critical processing of the advertisement.

The proposed path model visualizing our hypotheses can be found in Figure 1.

Figure 1

Proposed path model



Method

The study was approved by the ethics committee of Radboud University under approval number ECSW-LT-BC-2023-3-30-47641. We preregistered this study before collecting data: https://osf.io/mpurd. Every participant gave written informed consent before participation. Supplementary materials and our measures are publicly accessible on OSF: https://osf.io/zkjcb/?view_only=6d884a3377534df4a83151314c76cc28.

Design

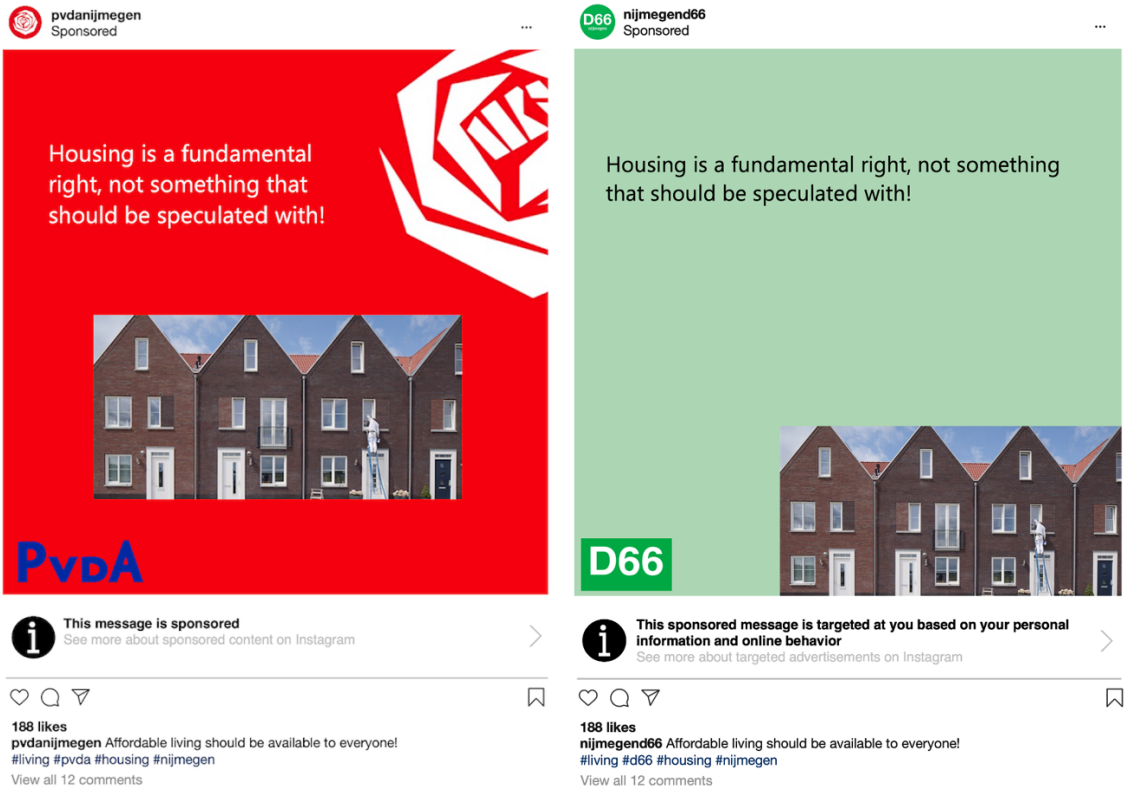
To test our hypotheses we conducted an eye-tracking laboratory experiment with a factorial between-subjects design containing two groups. Participants were exposed to an Instagram timeline containing six Instagram posts in total: three posts from Dutch news outlets, a post from a local hotel/restaurant, the university (which were all filler stimuli), and subsequently from a Dutch political party which contained either a sponsored disclosure (control condition) or a targeting disclosure (experimental condition) which can be found in Figure 2. The political advertisement was the third post in the timeline. The image of the full timelines can be found on OSF.

In both conditions, participants were exposed to the disclosure that Meta uses on Instagram to inform users that content is sponsored ('Sponsored', between the ad and account name). Moreover, underneath the advertisement, users were exposed to a more prominent disclosure, designed based on the disclosures Meta used in the COVID-19 pandemic. This disclosure underneath the post contained an "I" to show users that there was extra information to be found there. For participants in the control condition, the disclosure explained that the content was sponsored (in line with the term "sponsored" that was already visible above the ad). However, in the experimental condition, the disclosure contained additional information about the targeting practices taking place on the platform and that the ad was targeted at the users based on their personal information and online behavior. Thus, in both conditions a sponsorship disclosure was visible above the ad, while in only the experimental condition additional targeting information was presented. To ensure that possible effects of the manipulation would be due to the targeting disclosure and not just any text or disclosure being visible underneath the advertisement, in the control condition the sponsorship disclosure was repeated underneath the ad (See Figure 2).

To ensure that most of our participants at least could have been targeted by this advertisement in a real-world setting, we used the same political advertisement from two Dutch parties, one more center-left (Partij van de Arbeid; here "Labor") and one more center-right (Democrats 66; here "D66"), participants got shown one of the advertisements through random selection. Besides, since we predicted that the majority of our sample would be students and we aimed to make the ad about a relevant topic, the topic of the ad was the housing market, which is one of the most important political issues in the Netherlands (Statista, 2023) and stated: 'Housing is a fundamental right, not something that should be speculated with'. The ad was designed in the corporate design of both parties and contained a logo and the same picture of houses, we replicated the design from two local bodies of both political parties that we found through the Facebook Ad Library (Labor: https://www.facebook.com/ads/library/?id=122486037322954, D66: https://www.facebook.com/ads/library/?id=1240248669857229). The advertisement for the Labor Party also included a part of their larger logo (a rose), which was not the case for D66 since their logo is just the abbreviation of their name. On both Instagram posts, the number of likes (188) and comments (12) was the same, the ads differed from regular content regarding the timestamp of posting, which Instagram does not provide for ads.

Figure 1

Stimuli: Instagram posts for both the control condition (left) and targeting disclosure condition (right) for both political parties.



Procedure

Participants were invited to the laboratory and were informed that their eye movements and visual fixation would be recorded while they were looking at social media content. After calibrating the eye-tracker, we informed participants that they would see an Instagram timeline that they should browse ordinarily and that they could exit the timeline by pressing a button at the bottom of the screen. After the instruction, the Instagram timeline appeared, and visual fixation was recorded while the participants scrolled through the timeline. Next, participants were asked to fill in a questionnaire containing questions to measure the dependent variable and their demographic information. Finally, we debriefed and thanked our participants. The average time of completion for viewing the stimuli and answering the questions was 2 minutes and 19 seconds in total.

Apparatus

Visual fixation was recorded with a Tobii TX 300 eye-tracker (sampling rate of 300Hz). Participants were seated 60–70cm away from the monitor. A 9-point calibration was

performed before the start of the experiment. We excluded participants that had a weighted gaze of 80 percent during stimulus reception. This is a method for determining the quality of a recording which consists of a ratio of the number of samples that the eye-tracker correctly identified divided by the attempts. Finally, 134 participants met the criteria and could be included in data analyses (average proportion of recorded weighted gaze data: $M = 95.01\%$, $SD = 3.85$).

Sample

We recruited 150 participants that had at least one active social media account, through the online participant system of the university or by approaching them on campus between the 18th of April and the 15th of May 2023. We excluded participants with a weighted gaze percentage of less than 80%, leaving us with 134 participants (see Procedure). Participants' age ranged between 18 to 35 years ($M = 22.73$, $SD = 2.96$). Of these participants, 85 identified as female, 45 as male, 3 as non-confirming, and 1 as a transgender male. Regarding education, most of our participants had a high school diploma ($n = 70$), 55 participants had a bachelor's degree, 8 had a master's degree and 1 participant had a vocational education diploma. Randomization checks showed no differences between our two experimental groups regarding age ($F(1, 132) = 0.05$, $p = .817$), gender ($\chi2(3, N = 134) = 1.46$, $p = .691$), or level of education ($\chi2(3, N = 134) = 7.50$, $p = .058$). Moreover, there were no differences in which version of the political advertisement (D66 or Labor) participants saw between our experimental groups ($\chi2(1, N = 134) = 0.48$, $p = .489$).

Power

Given the budget for this study, we were able to gather responses from 150 participants. As described, we only analyzed data from participants with a weighted gaze percentage above 80, leaving us with a final sample of 134 participants. To determine our statistical power, we used R (version 4.1.2; R Core Team, 2021) and the simsem package

(version 0.5-13; Pornprasertmanit et al., 2021). The power analysis aimed to set the smallest effect size that can be found with a recommended minimum power of 90% (Curran-Everett, 2017). The smallest effect size of interest (SESOI; Lakens et al., 2018) is complementary to and comparable with the alpha-level, i.e., a threshold for rejecting results. We calculated our SESOI while using a power level of 90%, an α level of .05, and a sample size of 134 participants in total. Consequently, we set our SESOI at $\beta = |.28|$ and do not interpret results with effect sizes below that.

Measures

In this study, we measured both visual attention to the disclosure and visual attention to the ad in seconds, with the eye-tracker. To measure attention to the disclosure two areas of interest (AOIs) were combined into a measurement for visual attention to the disclosure in seconds. To measure attention to the ad we did the same thing but with different AOIs. We combined participants' visual fixation on the account name, the image, and the caption below the image into a measure for visual attention to the ad in seconds. Finally, as a measure for the critical processing of the ad, we adapted two items that have been used in targeting research by Zarouali and colleagues (2018) but originally was used in disclosure research by Boerman and colleagues (2014) on a Likert scale ranging from 1 (= strongly disagree) to 7 (= strongly agree): *"While scrolling the Instagram timeline I criticized the advertisement of the political party"*, and *"While scrolling the Instagram timeline I was skeptical toward the advertisement of the political party"*. We computed the mean for these two items as our measure for critical processing (Cronbach's $\alpha = .76$, McDonald's $\omega = .76$).

Manipulation check

As a manipulation check, we asked participants if they recalled if there was a disclosure on the Instagram ad and what the disclosure stated by asking them to check one of the following

statements: *"The Instagram post was a regular post"*, *"The Instagram post was labeled as targeted at me and sponsored"*, or *"The Instagram post was labeled as sponsored"*.

Besides, as a stimulus check, we asked participants if they thought that this advertisement could have been shown to them while browsing a SNS in real life on a Likert scale ranging from 1 (= definitely not) to 5 (= definitely yes) (M = 3.42, SD = 1.15).

Results

We conducted the statistical analyses for this work using R (version 4.1.2; R Core Team, 2021) and jamovi (version 2.3; The jamovi project, 2022). To investigate differences between our conditions we conducted a path model with our measures through structural equation modeling in lavaan (version 0.6-12.; Rosseel, 2012). The code that we used in both software packages is available on the OSF: https://osf.io/zkjcb/?view_only=6d884a3377534df4a83151314c76cc28. Means, standard deviations, and bivariate correlations for our measured variables can be found in Table 1. Descriptives per advertisement per group can be found on OSF. Other than preregistered, we changed the wording for our hypotheses to non-causal language (i.e., changed from lead to, to be related with).

Table 2

Means, Standard Deviations, and Bivariate Correlations of the Measured Constructs.

Measured construct	*M* (*SD*)	1	2	3	4
1 Experimental condition	-	-			
2 Visual attention to disclosure in *s*	2.20 (2.47)	.24**	-		
3 Visual attention to ad in *s*	6.69 (4.33)	-.00	.59***	-	
4 Critical processing	3.53 (1.46)	0.09	.37***	.19*	-

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Manipulation check

Concerning our manipulation check, in our control condition (n = 67) where the Instagram ad contained a sponsorship disclosure, 40 participants correctly recalled this disclosure, while 21 participants did not recall a disclosure at all and 6 participants wrongfully recalled a targeting disclosure. In our experimental condition (n = 67) where the Instagram ad contained a targeting disclosure, 25 participants recalled the correct disclosure, while 25 other participants wrongfully recalled a sponsored disclosure and 17 participants did not recall a disclosure at all. An overview of the recallment per experimental condition can be found in Table 3.

Table 3

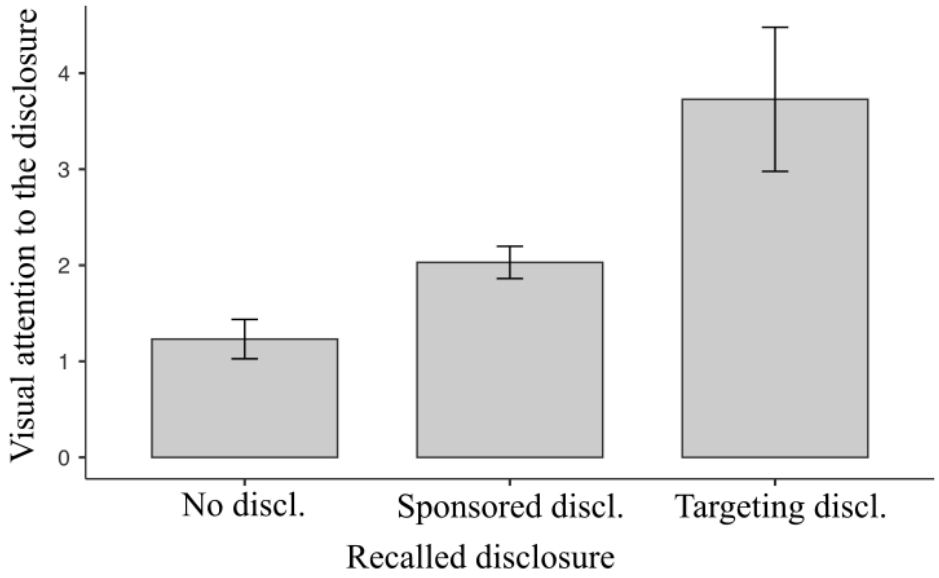
Results of the manipulation check for both conditions.

	Experimental condition	
Recalled disclosure	Control (sponsored) (n = 67)	Targeting disclosure (n = 67)
Sponsored	40	25
Targeting	6	25
No disclosure	21	17

Interestingly, we were able to investigate the recalled disclosure and participants' visual attention to the disclosure. We found that participants that recalled no disclosure, had the lowest visual attention in seconds to the disclosure (M = 1.23, SD = 1.26), followed by participants that recalled a sponsored disclosure (M = 2.03, SD = 1.35), and finally that participants who recalled a targeting disclosure, spent the most time looking at the disclosure (M = 3.37, SD = 4.18) a visual overview of these results can be found in Figure 2.

Figure 2

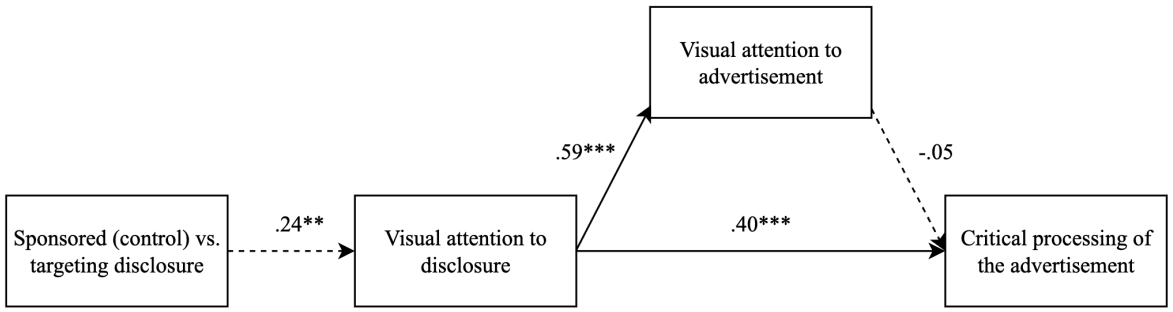
Visual attention to the disclosure and recalled disclosure.



As preregistered, we tested all our hypotheses in an integrated path model. We evaluated our model fit in line with frequently used fit indices (Hooper et al., 2008). Our model showed a near decent fit: $\chi2(2) = 4.57$, $p = .102$, $\chi2/df = 2.29$, CFI = 0.97, TLI = 0.91, RMSEA = .10, 90% CI [.00, .22], SRMR = .04. Our path model is shown in Figure 3.

Figure 3

Path Model



Note. Numbers represent standardized regression coefficients. Dashed lines indicate a path is either statistically meaningless (not significant) or theoretically meaningless (below $\beta = |.28|$). * $p < .05$, ** $p < .01$, *** $p < .001$.

For our first hypothesis, we expected that if an Instagram ad contained a targeting, this would affect participants' visual attention to the disclosure. Our results show a significant positive effect ($\beta = .24$, $p = .004$). However, this effect size is below our SESOI ($\beta = |.28|$), and thus, we are not able to interpret this effect, leading us to reject our first hypothesis. For our second hypothesis, we predicted that visual attention towards the disclosure would be positively related to users' critical processing of the message. We found proof for this in our data ($\beta = .40$, $p < .001$), leading us to accept this hypothesis. For our third hypothesis, we expected that visual attention towards the disclosure, would be positively related to visual attention towards the ad itself. We found proof for this in our data, ($\beta = .59$, $p < .001$), leading us to accept this hypothesis.

Finally, for our fourth hypothesis, we predicted that visual attention to the disclosure would be positively related to users' visual attention to the ad and that this, in turn, would positively relate to users' critical processing of the ad. We investigated this through mediation analyses in our path model in lavaan (version 0.6-12; Rosseel, 2012) while bootstrapping with 5,000 samples. We found no proof for mediation without our data as our indirect effect was not significant (indirect effect = -0.02, *SE* = 0.04, 95% *BCBCI* [-0.09, 0.05]), leading us to reject hypothesis four.

Discussion

In this work, we aimed to investigate whether a more informative targeting disclosure on a political microtargeted ad on Instagram would lead to more attention to the disclosure, if this led to more critical processing, and if this effect gets mediated by attention to the ad itself. To do so, we investigated targeting disclosures through a laboratory eye-tracking experiment in which we showed participants an Instagram timeline containing a post from a

political party containing either a sponsored or a targeting disclosure. Subsequently, we tested our hypotheses in an integrative path model before testing our mediation hypothesis through bootstrapping.

Firstly, we expected that exposure to a targeting disclosure, compared to a sponsored disclosure in our control condition, would lead to more attention to the disclosure (Binford et al., 2021; Boerman et al., 2015). Even though prior research shows mixed results concerning users' attention to a disclosure, we found a significant effect of our targeting disclosure on attention to this disclosure. However, the effect size that our data showed, was just below our smallest effect size of interest and thus, we were not able to meaningfully interpret this result.

Within existing disclosure research, results regarding the correct recallment of the disclosure are very diverse (Binder et al., 2022; Evans et al., 2017; Jost et al., 2022; Kruikemeier et al., 2016). The current study found the same results with regard to recallment, in our control condition with the sponsored disclosure, 60% of our participants recalled the correct disclosure, while in the experimental condition, 37% of our participants recalled the correct disclosure, and the same number of participants incorrectly recalled a sponsored disclosure. However, if we combine this information with the visual attention to the disclosure, we find that participants who recalled the targeting disclosure had 66% more fixation time on the disclosure compared to participants that recalled the sponsored disclosure. Besides, we found that participants that recalled the sponsored disclosure had 65% more fixation time than participants that recalled no disclosure. Even though it was not a core question in this study, we still think that the finding that 74% of our participants at least recalled a disclosure, is a step forward compared to other work (Binford et al., 2021; Jost et al., 2022; Kruikemeier et al., 2016). Especially since this is one of the first times that the disclosure was based on the COVID-19 disclosure that Meta used in times of the pandemic. Moreover, the finding that the same number of people recalled a sponsored disclosure in the

targeting condition and that 60% of people recalled the correct disclosure in the control condition, might imply that users need some time to adapt to this new type of disclosure and that eventually, they might internalize and recall it better, as they now did with the sponsored disclosure.

Besides the effect of our disclosure type on attention to this disclosure, we investigated if attention to the disclosure would lead to users engaging in more critical processing of the ad. In line with other work, we indeed found that within our sample, users that had more visual fixation on the disclosure, engaged in more critical processing of the advertisement (Boerman et al., 2014; Zarouali et al., 2017). Even though our first results that is described above, cannot be meaningfully interpreted within this study, we still think that the disclosure had an effect. Our line of thought behind this is that even without a correct recallment of disclosure that states the ad is targeted, the majority of participants still recalled a disclosure. Therefore, the disclosure could have been effective in helping users distinguish the ad from regular content. This could have helped them be aware of the persuasive attempt (Friestad and Wright, 1994), and subsequently manage their response by evaluating the ad systematically and critically (Boerman et al., 2014; Janssen et al., 2010).

Furthermore, we expected that attention to the disclosure would cause a backfire effect where it would lead to more attention to the advertisement. We indeed found this effect in our data and thus can say that disclosures might have a backfire effect (Janssen et al., 2016). This backfire effect can be an unintentional effect of disclosures where the label, besides protecting users from the persuasive effects of the ad, increases rather than decreases the persuasive effects. This is in line with existing work that shows that viewers' recognition of an ad increases their brand memory (Boerman et al., 2015). Besides, other research shows that sponsorship transparency can have a positive effect on receivers' attitude towards the advertisement (Evans et al., 2019). Moreover, in privacy and e-commerce research, a

disclosure about privacy has been shown to lead to more trust in a retailer (Pan and Zinkhan, 2006). Finally, earlier work on microtargeting shows that users' awareness of the persuasive meaning and the personalization taking place on the platform, positively influences party evaluations (Binder et al., 2022). A reason for this backfire effect might be that users are curious about the microtargeting, in a skeptical way that is different from critical processing of the ad. This might imply that users, if they are made aware of the persuasive nature of the ad as well as the targeting that took place to deliver that specific ad to them via their timeline, have a surveillance motive and are curious about the advertisement (Choi, 2016; Flavián and Gurrea, 2009). This could even lead to them allocating more cognitive resources toward the advertisement because they feel like they are researching what the platform thinks it knows about them as an individual.

Regarding our final hypothesis, we expected a mediating effect of attention to the ad on the effect of attention to the disclosure on critical processing. However, we did not find this mediating effect of attention to the ad in our data. This could imply that even though we think that people could be curious about the advertisement that is shown to them because they are informed that it is a persuasive message that is targeted to them, this might not lead them to be more critical about it after they spend some seconds on the advertisement. We found these results interesting since we did find relatively strong effects between attention to the disclosure and both critical processing and attention to the ad. However, this result aligns with the results from our path model that there was no relation between visual attention to the ad and critical processing of the ad.

Limitations and future work

In this work, we aimed to randomize the political ads participants saw by using a center-left and center-right party out of the Dutch party system. Besides, we choose the statement about the housing market to at least try to make the ad suitable for a sample that

would be mainly students. Even though our randomization check did not show any differences between our two conditions, this does not specifically mean that the content was targeted at the participants. The field of microtargeting research would benefit from more studies that use large-scale field experiments with actual usage data and existing advertisements, to improve ecological validity. This data and these ads could be gathered through web scraping or even data donations.

Another limitation concerning eye-tracking is that in the current study, we had an initial sample size of 150 participants that mainly were students. While this is more than the average of 73 participants and two-thirds being students, in studies published in the top 25 communication journals between 2005 and 2019 (King et al., 2019), future research could find different results with larger sample sizes.

Moreover, we would like to emphasize that not all our participants recalled the right disclosure. While most participants did notice a disclosure, which is something that not always happens in disclosure research, future work would benefit the field, as well as legislators and regulators, by taking a more practical approach and scrutinizing more precisely what factors influence the correct recalling of disclosure by focusing more on the design and the different elements that a disclosure could contain.

Moreover, we would like to point out that this study was carried out in a laboratory and that this could have influenced or overestimated our results. Participants in a laboratory are very aware of the fact that they take part in a study, especially when you tell them their eye movements and fixations are being measured. However, while eye-tracking studies enable us to gain valuable insights into participants' attention distribution, it is also a possibility that users perceive parts of the stimuli through peripheral vision without actually fixating on an area of interest. Finally, the timeline that participants saw, was not their own,

which might lead to an even stronger consciousness that they were in a laboratory viewing a made-up timeline.

Conclusion

Political microtargeting is something that users on social media do not actively recognize in most cases. Nevertheless, regulatory bodies, legislative authorities, and researchers alike agree that there is a need for improving transparency. The current study investigated the effects of targeting disclosures on attention to these disclosures and the effects of attention to the disclosure on critical processing of, and attention to, the ad. Finally, we investigated the potential mediation of attention to the ad on the effect of attention to the disclosure and critical processing. The results show that a more prominent disclosure was not related to more attention to the disclosure. But that visual attention, in turn, was positively related to both critical processing and attention to the displayed ad. Finally, attention to the ad did not relate to critical processing and attention to the ad did not mediate the relation between attention to the disclosure and critical processing of the ad. Together, these findings show that there is a potential for newly designed and more prominent disclosures, and that these would be able to inform users about the targeting taking place while obliging with the Digital Services Act. However, users that see the disclosure, might process an ad more critically, which could be fruitless for political advertisers.

Research Transparency Statement

The authors are willing to share their data, analytics methods, and study materials with other researchers. The material will be available at OSF.

Preregistration Statement

The authors have pre-registered this research with an analysis plan which is retrievable at: https://osf.io/mpurd/?view_only=0cec88db924245fd88d1255922417946

References

Aguirre, E., Mahr, D., Grewal, D., de Ruyter, K., and Wetzels, M. (2015). Unraveling the personalization paradox: The effect of information collection and trust-building strategies on online advertisement effectiveness. *Journal of Retailing*, *91*(1), 34–49. https://doi.org/10.1016/j.jretai.2014.09.005

Allyn, J., and Festinger, L. (1961). The effectiveness of unanticipated persuasive communications. *The Journal of Abnormal and Social Psychology*, *62*(1), 35–40. https://doi.org/10.1037/h0048493

Amazeen, M. A., and Wojdynski, B. W. (2020). The effects of disclosure format on native advertising recognition and audience perceptions of legacy and online news publishers. *Journalism*, *21*(12), 1965–1984. https://doi.org/10.1177/1464884918754829

Bennett, C. J. (2015). Trends in Voter Surveillance in Western Societies: Privacy Intrusions and Democratic Implications. *Surveillance & Society*, *13*(3/4), 370–384. https://doi.org/10.24908/ss.v13i3/4.5373

Binder, A., Stubenvoll, M., Hirsch, M., and Matthes, J. (2022). Why Am I Getting This Ad? How the Degree of Targeting Disclosures and Political Fit Affect Persuasion Knowledge, Party Evaluation, and Online Privacy Behaviors. *Journal of Advertising*, 1–17. https://doi.org/10.1080/00913367.2021.2015727

Binford, M. T., Wojdynski, B. W., Lee, Y.-I., Sun, S., and Briscoe, A. (2021). Invisible transparency: Visual attention to disclosures and source recognition in Facebook political advertising. *Journal of Information Technology & Politics*, *18*(1), 70–83. https://doi.org/10.1080/19331681.2020.1805388

Bodó, B., Helberger, N., and De Vreese, C. H. (2017). Political micro-targeting: A candidate or just a dark horse? *Internet Policy Review*, *6*(4). https://doi.org/10.14763/2017.4.776

Boerman, S. C., and Müller, C. M. (2021). Understanding which cues people use to identify influencer marketing on Instagram: An eye tracking study and experiment. *International Journal of Advertising*, 1–24. https://doi.org/10.1080/02650487.2021.1986256

Boerman, S. C., and van Reijmersdal, E. A. (2016). Informing Consumers about "Hidden" Advertising: A Literature Review of the Effects of Disclosing Sponsored Content. In P. De Pelsmacker (Ed.), *Advertising in New Formats and Media* (pp. 115–146). Emerald Group Publishing Limited. https://doi.org/10.1108/978-1-78560-313-620151005

Boerman, S. C., van Reijmersdal, E. A., and Neijens, P. C. (2014). Effects of Sponsorship Disclosure Timing on the Processing of Sponsored Content: A Study on the Effectiveness of European Disclosure Regulations: EFFECTS OF SPONSORSHIP DISCLOSURE TIMING. *Psychology & Marketing*, *31*(3), 214–224. https://doi.org/10.1002/mar.20688

Boerman, S. C., van Reijmersdal, E. A., and Neijens, P. C. (2015). Using Eye Tracking to Understand the Effects of Brand Placement Disclosure Types in Television Programs. *Journal of Advertising*, *44*(3), 196–207. https://doi.org/10.1080/00913367.2014.967423

Cadwalladr, C. (2018, March 18). 'I made Steve Bannon's psychological warfare tool': Meet the data war whistleblower. *The Guardian*. https://www.theguardian.com/news/2018/mar/17/data-war-whistleblower-christopher-wylie-faceook-nix-bannon-trump

Campbell, M. C., and Kirmani, A. (2000). Consumers' Use of Persuasion Knowledge: The Effects of Accessibility and Cognitive Capacity on Perceptions of an Influence Agent. *Journal of Consumer Research*, *27*(1), 69–83. https://doi.org/10.1086/314309

Choi, J. (2016). Why do people use news differently on SNSs? An investigation of the role of motivations, media repertoires, and technology cluster on citizens' news-related activities. *Computers in Human Behavior*, *54*, 249–256. https://doi.org/10.1016/j.chb.2015.08.006

Clark, M. (2020, December 17). *Instagram is rolling out new notifications about COVID-19 information*. The Verge. https://www.theverge.com/2020/12/17/22187298/instagram-coronavirus-covid19-misinformation-notifications

Curran-Everett, D. (2017). CORP: Minimizing the chances of false positives and false negatives. *Journal of Applied Physiology*, *122*(1), 91–95. https://doi.org/10.1152/japplphysiol.00937.2016

Dobber, T. (2021). *Effectiveness of Online Political Ad Disclosure Labels: Empirical Findings*. 6.

Dobber, T., Kruikemeier, S., Helberger, N., and Goodman, E. (2023). Shielding citizens? Understanding the impact of political advertisement transparency information. *New Media & Society*.

Dobber, T., Ó Fathaigh, R., and Zuiderveen Borgesius, F. J. (2019). The regulation of online political micro-targeting in Europe. *Internet Policy Review*, *8*(4). https://doi.org/10.14763/2019.4.1440

European Commission. (2022). *REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on a Single Market For Digital Services (Digital Services Act) and amending Directive*. https://eur-lex.europa.eu/legal-content/en/TXT/?uri=COM:2020:825:FIN

Evans, N. J., Phua, J., Lim, J., and Jun, H. (2017). Disclosing Instagram Influencer Advertising: The Effects of Disclosure Language on Advertising Recognition, Attitudes, and Behavioral Intent. *Journal of Interactive Advertising*, *17*(2), 138–149. https://doi.org/10.1080/15252019.2017.1366885

Evans, N. J., Wojdynski, B. W., and Grubbs Hoy, M. (2019). How sponsorship transparency mitigates negative effects of advertising recognition. *International Journal of Advertising*, *38*(3), 364–382. https://doi.org/10.1080/02650487.2018.1474998

Lutte contre la manipulation de l'information, 2018, 1202, 0297 (2018). https://www.senat.fr/dossier-legislatif/ppl17-623.html

Flavián, C., and Gurrea, R. (2009). Users' motivations and attitude towards the online press. *Journal of Consumer Marketing*, *26*(3), 164–174. https://doi.org/10.1108/07363760910954109

Friestad, M., and Wright, P. (1994). The Persuasion Knowledge Model: How People Cope with Persuasion Attempts. *Journal of Consumer Research*, *21*(1), 1–31. https://doi.org/10.1086/209380

Gandy, O. H. (2000). Dividing Practices: Segmentation and Targeting in the Emerging Public Sphere. In W. L. Bennett and R. M. Entman (Eds.), *Mediated Politics* (1st ed., pp. 141–159). Cambridge University Press. https://doi.org/10.1017/CBO9780511613852.008

Giasson, T., Le Bars, G., and Dubois, P. (2019). Is Social Media Transforming Canadian Electioneering? Hybridity and Online Partisan Strategies in the 2012 Quebec Election. *Canadian Journal of Political Science*, *52*(2), 323–341. https://doi.org/10.1017/S0008423918000902

Holt, K., Shehata, A., Strömbäck, J., and Ljungberg, E. (2013). Age and the effects of news media attention and social media use on political interest and participation: Do social

media function as leveller? *European Journal of Communication*, *28*(1), 19–34. https://doi.org/10.1177/0267323112465369

Hooper, D., Couglan, J., and Mullen, M. R. (2008). Structural equation modelling: Guidelines for determining model fit. *Journal of Business Research Methods*, *6*(1), 53–60.

Instagram. (2020, March 16). *Helping to Inform People about COVID-19 Vaccines | Instagram Blog*. https://about.instagram.com/blog/announcements/continuing-to-keep-people-safe-and-informed-about-covid-19

Jamieson, K. H. (2013). Messages, micro-targeting, and new media technologies. *Forum (Germany)*, *11*(3), 429–435. https://doi.org/10.1515/for-2013-0052

Janssen, L., Fennis, B. M., and Pruyn, A. Th. H. (2010). Forewarned is forearmed: Conserving self-control strength to resist social influence. *Journal of Experimental Social Psychology*, *46*(6), 911–921. https://doi.org/10.1016/j.jesp.2010.06.008

Janssen, L., Fransen, M. L., Wulff, R., and van Reijmersdal, E. A. (2016). Brand placement disclosure effects on persuasion: The moderating role of consumer self-control: Self-control and brand placement disclosure. *Journal of Consumer Behaviour*, *15*(6), 503–515. https://doi.org/10.1002/cb.1587

Janssen, L., Schouten, A. P., and Croes, E. A. J. (2021). Influencer advertising on Instagram: Product-influencer fit and number of followers affect advertising outcomes and influencer evaluations via credibility and identification. *International Journal of Advertising*, 1–27. https://doi.org/10.1080/02650487.2021.1994205

Jost, P., Kruschinski, S., Sülflow, M., Haßler, J., and Maurer, M. (2022). Invisible transparency: How different types of ad disclaimers on Facebook affect whether and how digital political advertising is perceived. *Policy & Internet*, poi3.333. https://doi.org/10.1002/poi3.333

King, A. J., Bol, N., Cummins, R. G., and John, K. K. (2019). Improving Visual Behavior Research in Communication Science: An Overview, Review, and Reporting Recommendations for Using Eye-Tracking Methods. *Communication Methods and Measures*, *13*(3), 149–177. https://doi.org/10.1080/19312458.2018.1558194

King, J. M. (2022). Microtargeted Political Ads: An Intractable Problem. *Boston University Law Review*, *102*, 39.

Kruikemeier, S., Sezgin, M., and Boerman, S. C. (2016). Political Microtargeting: Relationship between Personalized Advertising on Facebook and Voters' Responses. *Cyberpsychology, Behavior, and Social Networking*, *19*(6), 367–372. https://doi.org/10.1089/cyber.2015.0652

Kruikemeier, S., Vermeer, S., Metoui, N., Dobber, T., and Zarouali, B. (2022). (Tar)getting you: The use of online political targeted messages on Facebook. *Big Data & Society*, *9*(2), 20539517221089626. https://doi.org/10.1177/20539517221089626

Lakens, D. (2022). Sample Size Justification. *Collabra: Psychology*, *8*(1), 33267. https://doi.org/10.1525/collabra.33267

Lakens, D., Scheel, A. M., and Isager, P. M. (2018). Equivalence Testing for Psychological Research: A Tutorial. *Advances in Methods and Practices in Psychological Science*, *1*(2), 259–269. https://doi.org/10.1177/2515245918770963

Online Advertising and Social Media (Transparency) Bill, no. 150–2017 (2021). https://data.oireachtas.ie/ie/oireachtas/bill/2017/150/eng/initiated/b15017d.pdf

Lubianco, J. (2022, April 12). *Fake News bill gets stuck in Brazilian Congress and it's unlikely to be voted on before the elections; remuneration proposal for journalistic organizations is a sensitive topic*. LatAm Journalism Review by the Knight Center. https://latamjournalismreview.org/articles/fake-news-brazil-payment-journalism/

Matthes, J., Hirsch, M., Stubenvoll, M., Binder, A., Kruikemeier, S., Lecheler, S., and Otto, L. (2022). Understanding the democratic role of perceived online political micro-targeting: Longitudinal effects on trust in democracy and political interest. *Journal of Information Technology & Politics*, 1–14. https://doi.org/10.1080/19331681.2021.2016542

Murray, G. R., and Scime, A. (2010). Microtargeting and electorate segmentation: Data mining the American National Election Studies. *Journal of Political Marketing*, *9*(3), 143–166. https://doi.org/10.1080/15377857.2010.497732

Pan, Y., and Zinkhan, G. M. (2006). Determinants of retail patronage: A meta-analytical perspective. *Journal of Retailing*, *82*(3), 229–243. https://doi.org/10.1016/j.jretai.2005.11.008

Pornprasertmanit, S., Miller, P., Schoemann, A., Jorgensen, T. D., and Quick, C. (2021). *simsem: SIMulated Structural Equation Modeling* (0.5-16). https://CRAN.R-project.org/package=simsem

R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Rosseel, Y. (2012). **lavaan**: An *R* Package for Structural Equation Modeling. *Journal of Statistical Software*, *48*(2). https://doi.org/10.18637/jss.v048.i02

Tene, O. (2011). Privacy: The new generations. *International Data Privacy Law*, *1*(1), 15–27. https://doi.org/10.1093/idpl/ipq003

The jamovi project. (2022). *Jamovi* (2.3). http://www.jamovi.org

Wojdynski, B. W., and Evans, N. J. (2016). Going Native: Effects of Disclosure Position and Language on the Recognition and Evaluation of Onlin. *Journal of Advertising*, *45*(2), 157–168. https://doi.org/10.1080/00913367.2015.1115380

Zarouali, B., Dobber, T., De Pauw, G., and de Vreese, C. (2020). Using a Personality-Profiling Algorithm to Investigate Political Microtargeting: Assessing the Persuasion Effects of Personality-Tailored Ads on Social Media. *Communication Research*. https://doi.org/10.1177/0093650220961965

Zarouali, B., Poels, K., Ponnet, K., and Walrave, M. (2018). "Everything under control?": Privacy control salience influences both critical processing and perceived persuasiveness of targeted advertising among adolescents. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace*, *12*(1). https://doi.org/10.5817/CP2018-1-5

Zarouali, B., Ponnet, K., Walrave, M., and Poels, K. (2017). "Do you like cookies?" Adolescents' skeptical processing of retargeted Facebook-ads and the moderating role of privacy concern and a textual debriefing. *Computers in Human Behavior*, *69*, 157–165. https://doi.org/10.1016/j.chb.2016.11.050

Zuiderveen Borgesius, F. J., Möller, J., Kruikemeier, S., Fathaigh, R., Irion, K., Dobber, T., Bodo, B., and de Vreese, C. (2018). Online political microtargeting: Promises and threats for democracy. *Utrecht Law Review*, *14*(1), 82–96. https://doi.org/10.18352/ulr.420